Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Music-Aware Agentic AI Remix System and Reasoning-Based Evaluation Framework**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication			
	Medium Category	Small Category	Medium Category	Small Category
	Digital Content	*Digital Content Production and Distribution*		
Concept of Proposed Technology	**Core Concept** The proposed technology is a Music-Aware Agentic AI Remix System combined with a Reasoning-Based Evaluation Framework, designed to enable AI-generated remixing that is musically coherent, structurally interpretable, and attribution-safe. **Key Functions** Unlike conventional generative music models that treat remixing as a purely signal-level transformation, this system approaches remixing as a structured musical decision-making process executed by multiple specialized AI agents. Each agent is responsible for a distinct musical or operational role, including: 1) Musical structure analysis (form, sections, phrases, blocks) 2) Music-aware remix decision-making and transformation planning 3) Generative synthesis and controlled variation			

	4) Block-level attribution, traceability, and usage tracking of original musical assets A central design principle of the system is that AI remixing should exhibit reasoning aligned with musical structure and intent, rather than producing unconstrained or opaque outputs. To support this goal, the project introduces a Reasoning-Based Evaluation Framework that formalizes how musical intelligence in AI remixing can be assessed. This framework encodes expert musical reasoning into structured evaluation logic, enabling systematic analysis of whether agent decisions align with principles of musical coherence, form, and expressive intent. **Potential Applications** The proposed technology is applicable across a wide range of music creation and distribution scenarios, including: 1) Fan and Creator Remix Platforms: Enabling users to create AI-assisted remixes of existing songs while preserving musical structure and respecting artist rights through transparent attribution. 2) AI-Assisted Music Production Tools: Supporting composers and producers with intelligent remixing, variation exploration, and structure-aware creative assistance. 3) Personalized and Interactive Music Services: Powering adaptive remixes for personalized playlists, AI radio, and interactive listening experiences. 4) Licensable AI Music Infrastructure: Providing music platforms, distributors, and rightsholders with remix-capable AI systems that are both musically interpretable and legally viable. 5) Evaluation and Benchmarking of Music AI Systems: Applying the reasoning-based evaluation framework as a reusable methodology to assess musical intelligence in other AI music generation and transformation systems. Through these applications, the technology aims to make AI-driven remixing musically meaningful, explainable, and scalable for global markets.
Rationale for Joint R&D Project	**Critical Technical Barriers** Despite recent advances in AI music generation, there is no established methodology for evaluating whether AI systems truly understand and respect musical structure during remixing. Most existing evaluation approaches rely on signal-level metrics or subjective listening tests, which are insufficient for assessing musical reasoning and decision-making. **Why Global Collaboration?** Collaboration with an internationally recognized music technology research partner such as Tae Hong Park is essential to: 1) Ground the system's agent behavior in established musical principles, 2) Translate expert musical reasoning into structured evaluation models, and 3) Validate the system through credible, external expert assessment. Global collaboration is essential for this project for the following reasons: 1) Concentration of Specialized Expertise: Music-aware AI evaluation lies at the intersection of music theory, audio signal processing, and AI system design—expertise that is globally

	distributed and rarely available within a single organization or country. 2) Objective and Credible Validation: International collaboration ensures neutral, academically rigorous evaluation, enhancing the global credibility and transferability of the research outcomes. 3) Global Nature of Music AI Services: AI remix systems operate across international music catalogs, platforms, and audiences. Global validation is therefore necessary to ensure broad applicability and acceptance. 4) Foundation for Scalable International R&D: This evaluation-oriented joint research establishes a foundation for future large-scale international R&D phases, including system deployment, standardization, and commercialization. This challenge cannot be adequately addressed through algorithmic performance metrics alone. It requires expert evaluation and theoretical grounding from the field of music technology and music structure analysis, particularly in areas such as: 1) Structural coherence across sections and phrases 2) Musical plausibility of transitions and transformations 3) The balance between rule-based musical knowledge and learned generative behavior 4) Perceptual and analytical judgments of “musically intelligent” remix decisions Therefore, collaboration with a specialized research partner in music technology and audio-based musical intelligence is essential. In particular, evaluation and guidance from a domain expert such as Tae Hong Park would provide authoritative insight into whether the system’s agentic behavior aligns with established principles of musical structure and interpretation. Through this joint R&D project, the objective is not only to improve a specific AI system, but to establish a research framework for music-aware agentic remixing, contributing to the global discourse on how AI can engage with music as a structured and meaningful creative domain. **Expected Impact** This project addresses that gap by jointly developing: 1) A music-aware Agentic AI Remix System, and 2) A reasoning-based evaluation framework that formalizes expert musical judgment. Such an effort requires deep interdisciplinary collaboration between AI system developers and music technology researchers with expertise in musical structure, audio analysis, and human musical reasoning.
Key Words	1. Agentic AI 2. Music Technology 3. Reasoning-Based Evaluation 4. Music Structural Intelligence 5. Attribution-Safe Music Generation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	**R&D Objectives** The overarching objective of this R&D project is to develop, evaluate, and validate a music-aware Agentic AI Remix System together with a structured, reasoning-based evaluation framework that can assess musical intelligence beyond signal-level performance. **Detailed Objectives** 1) Develop Music-Aware Evaluation Criteria and Reasoning Frameworks: Define evaluation dimensions that capture musical structure, coherence, and intent, and formalize reasoning processes used to judge musical validity in AI remix outputs. 2) Design and Implement a Reasoning-Based Evaluation Model: Develop an evaluation model that combines expert musical reasoning, structured rules, and interpretable AI components to assess agentic remix decisions. 3) Evaluate Agent Decision-Making from a Musical and Cognitive Perspective: Analyze how remix agents reason about musical form, transitions, and variation, and whether these processes align with human musical judgment. 4) Identify and Formalize Musical and Structural Failure Modes: Systematically identify failure cases in remix outputs and encode them into an evaluation taxonomy usable for both expert review and automated assessment. 5) Refine Agent Architecture Using Evaluation Feedback: Improve agent orchestration, constraints, and decision logic based on insights derived from the evaluation framework. 6) Demonstrate a Validated Music-Aware Agentic Remix Pilot System: Validate both the system and the evaluation framework through expert-reviewed pilot demonstrations in realistic creative workflows.			
R&D Target Specifications	**Qualitative Targets** 1) Remix outputs preserve musical form and structural logic. 2) Agent decisions can be interpreted and explained through the evaluation framework. 3) Evaluation results are consistent with expert musical judgment. **Quantitative / Semi-Quantitative Targets** 1) AI Agent Expert Evaluation Agreement Rate: ≥ 80% agreement between expert judgments and reasoning-based evaluation model outputs. 2) Expert Evaluation Score: Average score of ≥ 4.0 / 5.0 on musical coherence and structure. 3) Failure Detection Coverage: ≥ 85% of identified musical failure cases detectable by the evaluation model. Reduction of Musical Failure Cases: ≥ 30% reduction compared to the baseline system.			

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R&D Plan	**Overall R&D Strategy** This R&D plan follows a three-year, evaluation-driven development strategy, in which: 1) Year 1 focuses on establishing music-aware reasoning and evaluation foundations, 2) Year 2 focuses on aligning evaluation models with agentic system behavior and refining both, and 3) Year 3 focuses on pilot-scale validation, operational demonstration, and readiness for commercialization-oriented R&D. The roles of the Korean SME and research institute are clearly separated yet tightly coupled to ensure effective collaboration. **Year 1: Music-Aware Evaluation Framework & Baseline System Analysis** 1) Objectives ● Establish a reasoning-based evaluation framework for music-aware AI remixing. ● Analyze the baseline Agentic AI Remix System from a musical intelligence perspective. 2) Key Deliverables ● Music-aware reasoning-based evaluation framework ● Baseline expert evaluation report ● Initial taxonomy of musical and structural failure modes **Year 2: Evaluation Model Alignment & Agentic System Refinement** 1) Objectives ● Align the reasoning-based evaluation model with agentic system behavior. ● Refine both the evaluation framework and remix agents based on expert feedback. 2) Key Deliverables (Year 2) ● Expert-aligned reasoning-based evaluation model ● Refined Agentic AI Remix System (evaluation-integrated version) ● Comparative evaluation results (baseline vs. refined system) **Year 3: Pilot-Scale Demonstration & Operational Validation** 1) Objectives ● Validate the system and evaluation framework in operational environments. ● Demonstrate readiness for commercialization-oriented R&D and global deployment. 2) Key Deliverables (Year 3) ● Pilot-scale Agentic AI Remix System (TRL 6) ● Final expert validation report ● Technical documentation for commercialization and follow-up R&D

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	Summary Through this three-year joint R&D plan, the project advances from a functional prototype (TRL 5) to an expert-validated pilot system (TRL 6) by jointly developing: 1) A music-aware Agentic AI Remix System, and 2) A reasoning-based evaluation framework grounded in expert musical intelligence. The clear division of roles between Korea SME partner and research institute —particularly under the guidance of Tae Hong Park, ensures both technical depth and academic credibility, forming a strong foundation for global collaboration and commercialization.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute, Purdue University, will serve as the lead research partner for music intelligence and reasoning-based evaluation, providing academic expertise in music technology, musical structure analysis, and expert-driven assessment methodologies. The expected roles and responsibilities include: ● Development of Music-Aware Evaluation Frameworks: Define evaluation dimensions for music-aware AI remixing, including musical form, structural coherence, section and phrase transitions, and perceived musical intent. ● Design of Reasoning-Based Evaluation Models: Formalize expert musical reasoning into structured evaluation logic, including rule-based, reasoning-driven, or interpretable model components that assess agentic remix decisions. ● Expert Evaluation and Validation: Conduct expert-led evaluation sessions on remix outputs, provide qualitative and semi-quantitative assessments, and validate whether agent behaviors align with established musical principles. ● Failure Mode Identification and Analysis: Identify, categorize, and document musical and structural failure cases in AI remix outputs, contributing to a reusable taxonomy for music-aware AI evaluation.
Expected R&R of Korean SME Partner	The Korean SME partner will act as the lead developer and system integrator, responsible for designing, implementing, and operating the Agentic AI Remix System and associated evaluation infrastructure. The expected roles and responsibilities include: ● Development of Agentic AI Remix System: Design and implement agent architectures for music structure analysis, remix decision-making, generative transformation, and block-level attribution. ● System Integration and Orchestration: Integrate individual agents into a cohesive, scalable system, including orchestration logic, data pipelines, and logging mechanisms required for evaluation. ● Implementation of Evaluation Pipelines: Implement automated or semi-automated evaluation workflows that incorporate the reasoning-based evaluation models developed in collaboration with research institute.

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	● System Refinement and Optimization: Refine agent behavior, constraints, and control policies based on evaluation feedback, and optimize system performance for pilot-scale operation. ● Pilot Deployment and Validation: Deploy the refined system in realistic creative and service-oriented environments, conduct pilot demonstrations, and validate system stability, usability, and scalability. ● Commercialization and Technology Transfer Preparation: Prepare documentation, demonstrations, and technical assets required for follow-up R&D, industry partnerships, and commercialization.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to generate significant technological, commercial, and industrial impact by advancing both AI-based music remix systems and evaluation methodologies for musical intelligence. **Technological Outcomes** ● Validated Music-Aware Agentic AI Remix System (TRL 6): A pilot-scale Agentic AI Remix System that demonstrates musically coherent, structure-aware remixing validated through expert evaluation and operational demonstrations. ● Reasoning-Based Evaluation Framework for Music AI: A structured evaluation methodology that formalizes expert musical reasoning, enabling systematic assessment of musical intelligence in AI remix systems beyond signal-level metrics. ● Explainable and Interpretable AI Remix Decisions: Improved transparency in agent decision-making, allowing remix behaviors to be interpreted, analyzed, and refined based on musical principles. ● Reusable Evaluation Assets: Evaluation criteria, failure-mode taxonomies, and reasoning models that can be reused or extended for other AI music generation and transformation systems. **Commercialization Potential** ● Foundation for Licensable AI Remix Services: The validated system enables commercially viable AI remix platforms that respect musical structure and attribution, supporting sustainable revenue models for rightsholders and platforms. ● Acceleration of Product Readiness: Achieving TRL 7 positions the technology for rapid transition to customer trials, strategic partnerships, and follow-up commercialization-oriented R&D. ● Differentiation in the Global Music AI Market: The integration of music-aware reasoning and evaluation provides a strong competitive advantage over black-box generative music systems. **Industrial and Economic Impact** ● Establishment of Evaluation Standards for Music AI: The reasoning-based evaluation framework contributes toward emerging global standards for assessing musical intelligence in AI systems. ● Expansion of the AI Music Ecosystem: By enabling musically coherent and explainable remixing, the project supports broader adoption of AI in music creation, production, and distribution. ● Strengthening Global Collaboration: The joint research framework strengthens long-term international collaboration between Korean SMEs and global research institutions, facilitating future large-scale R&D initiatives.

	● Contribution to Korea's AI and Digital Content Competitiveness: The project enhances Korea's leadership in AI-converged creative technologies and supports global market expansion for Korean AI music platforms. **Academic and Societal Impact** ● Knowledge Contribution to Music Technology Research: The project produces academically credible methodologies and insights into how AI systems can reason about musical structure and intent. ● Promotion of Responsible AI in Creative Domains: By emphasizing interpretability, attribution, and musical validity, the project contributes to responsible and ethical deployment of AI in the creative industries.

6. References

(1) From Generation to Attribution: Music AI Agent Architectures for the Post-Streaming Era, NeurIPS AI4Music Workshop, 2025, Neutune, Inc. (Jongpil Lee)

(2) Introduction to digital signal processing: Computer musically speaking, World Scientific, 2010, Tae Hong Park